Directed Containers as Categories and Their Morphisms as Cofunctors

Extracted from a machine-checked development

Abstract

We formalise the object-level dictionary between directed containers and small categories: every directed container presents a small category whose objects are shapes and whose morphisms are positions with specified codomain; every small category yields a directed container whose positions are outgoing morphisms. The two constructions are mutually inverse up to a pointwise bijection on positions, not up to definitional equality. At the morphism level, directed-container morphisms coincide with cofunctors of the presented categories: the two structures carry identical data, compose strictly, and the correspondence is an isomorphism of categories. The proofs of all stated results have been machine-checked; we record only the mathematical content.

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1 Introduction

Directed containers equip a container (S, P) with root, sub-shape, and shift operations satisfying five laws D1–D5. Reading the data categorically—shapes as objects, positions as morphisms, root as identity, shift as composition—one obtains a small category. Conversely, from a small category one builds a directed container by taking outgoing morphisms as positions. This is the object-level dictionary of Ahman and Uustalu.

The two constructions are inverse only up to isomorphism. Starting from a directed container D, the round trip replaces each position type $P(s)$ by the total space of the fibration of positions over their codomains—each position tagged with the shape it lands on and a proof of that landing. We exhibit the evident pointwise bijection and leave the upgrade to an isomorphism of directed containers for later work.

At the morphism level the situation is cleaner. A directed-container morphism has a covariant shape map and a contravariant position map, compatible with root, sub-shape, and shift. This is not a functor of the presented categories (the variance on hom-sets is wrong); it is a cofunctor. Under the dictionary the two structures are definitionally the same data, and the correspondence is an isomorphism of categories.

What this note claims. In ordinary mathematical language: (i) directed containers present small categories, and small categories present directed containers, inverse up to a canonical bijection on positions; (ii) morphisms of directed containers *are* cofunctors of those categories, and the identification is an isomorphism **DCont** $\cong$ **Cof**.

2 Preliminaries

We work in a single universe **Type**.

Definition 2.1 (Directed container)**.** A **directed container** D consists of a shape type S, a position family P, a root $\mathsf{o}(s) \in P(s)$, a sub-shape $s \downarrow p \in S$, and a shift $p \oplus q \in P(s)$ for $q \in P(s \downarrow p)$, subject to:

D1. $s \downarrow \mathsf{o}(s) = s$;

D4. $s \downarrow (p \oplus q) = (s \downarrow p) \downarrow q$;

D2. $\mathsf{o}(s) \oplus q = (\mathrm{D1}) \cdot q$ (transport along D1);

D3. $p \oplus \mathsf{o}(s \downarrow p) = p$;

D5. $(p \oplus q) \oplus \big((\mathrm{D4})^{-1} \cdot q'\big) = p \oplus (q \oplus q')$ (transport along D4).

Definition 2.2 (Small category)**.** A **small category** $\mathcal{C}$ consists of a type of objects, a type of morphisms $\mathrm{Hom}(a, b)$ between any two objects, identity morphisms, and composition in *diagrammatic* order

$$\mathrm{comp} : \mathrm{Hom}(a, b) \to \mathrm{Hom}(b, c) \to \mathrm{Hom}(a, c),$$

subject to the usual left unit, right unit, and associativity laws. Diagrammatic order matches the shift of a directed container: $p \oplus q$ first travels along p, then along q.

Lemma 2.3 (Transport cancellation)**.** *For a family P over a type α, an equality $h : x = y$ in α, and $p \in P(y)$,*

$$h \cdot (h^{-1} \cdot p) \;=\; p. \tag{1}$$

Proof. By lean. □

3 From directed containers to small categories

Lemma 3.1 (Codomain of a composite)**.** *Let D be a directed container, and let $a, b, c \in S$ with $p \in P(a)$, $h_p : a \downarrow p = b$, $q \in P(b)$, and $h_q : b \downarrow q = c$. Then*

$$a \downarrow \big(p \oplus (h_p^{-1} \cdot q)\big) \;=\; c. \tag{2}$$

Proof. By lean. □

Remark 3.2. This is D4 with both shape equalities made into hypotheses, so that the subtype property of composition in the induced category is well-typed.

Definition 3.3 (Presented category)**.** Let D be a directed container. The **presented small category** $\mathcal{C}(D)$ has

1. objects: the shapes S;
2. morphisms $a \to b$: the subtype $\{p \in P(a) \mid a \downarrow p = b\}$;
3. identity at a: $\langle \mathsf{o}(a),\, \mathrm{D1}_a \rangle$;
4. composition of $\langle p, h_p \rangle : a \to b$ with $\langle q, h_q \rangle : b \to c$:

$$\big\langle p \oplus (h_p^{-1} \cdot q),\ \text{the equality of Lemma 3.1} \big\rangle.$$

Theorem 3.4 (Category laws for $\mathcal{C}(D)$)**.** *The data of Definition 3.3 satisfy the left unit, right unit, and associativity laws of a small category. Explicitly:*

1. *left unit uses D2 and transport cancellation (Lemma 2.3);*
2. *right unit uses D3;*
3. *associativity uses D5.*

In each case one substitutes the codomain equalities, after which transports reduce definitionally (proofs of shape equalities are definitionally irrelevant), leaving the corresponding directed-container law.

Proof. By lean. □

4 From small categories to directed containers

Definition 4.1 (Presented directed container)**.** Let $\mathcal{C}$ be a small category. The **presented directed container** $D(\mathcal{C})$ has

1. shapes: the objects of $\mathcal{C}$;
2. positions over s: outgoing morphisms $\sum_{s'} \mathrm{Hom}_{\mathcal{C}}(s, s')$;
3. root at s: $\langle s,\, \mathrm{id}_s \rangle$;
4. sub-shape of $\langle s',\, f \rangle$: the codomain s';
5. shift of $\langle s',\, f \rangle$ by $\langle s'',\, g \rangle$: $\langle s'',\, g \circ f \rangle$ (diagrammatic composition in $\mathcal{C}$).

Theorem 4.2 (Directed laws for $D(\mathcal{C})$)**.** *The data of Definition 4.1 satisfy D1–D5. Laws D1 and D4 hold strictly (the codomain of a composite is the codomain of the second factor, definitionally), so the transports in D2 and D5 vanish; the remaining laws are exactly the category laws of $\mathcal{C}$.*

Proof. By lean. □

5 The object-level round trip

Starting from a directed container D, the composite $D \mapsto \mathcal{C}(D) \mapsto D(\mathcal{C}(D))$ is not definitionally D: its positions over s are

$$\sum_{s'} \{p \in P(s) \mid s \downarrow p = s'\},$$

each position tagged with its codomain and a proof. The total space of this fibration is in bijection with $P(s)$.

Definition 5.1 (Round-trip position maps)**.** Let D be a directed container and $s \in S$.

1. **Forward.** Send $p \in P(s)$ to $\langle s \downarrow p,\ \langle p,\ \mathsf{refl} \rangle \rangle$ in $D\big(\mathcal{C}(D)\big)$'s position type at s.
2. **Backward.** Send $\langle s',\ \langle p,\ h \rangle \rangle$ to the underlying position $p \in P(s)$.

Theorem 5.2 (Position bijection, forward-then-back)**.** *For every directed container D, shape s, and position $p \in P(s)$, applying the forward map and then the backward map returns p.*

Proof. By lean. □

Theorem 5.3 (Position bijection, back-then-forward)**.** *For every directed container D, shape s, and position q of $D\big(\mathcal{C}(D)\big)$ at s, applying the backward map and then the forward map returns q.*

Proof. By lean. □

Remark 5.4 (Honesty about the gap)**.** The pointwise bijection of Theorems 5.2 and 5.3 does not by itself yield a definitional equality of directed containers, nor automatically an isomorphism of directed containers: one would need a notion of directed-container isomorphism that transports root, sub-shape, and shift along the bijection. That upgrade, and the resulting equivalence of categories $\mathbf{DCont} \simeq \mathbf{Cat}$ (Ahman–Uustalu 2016, Theorem 3.6), remains at the paper level relative to the present formalisation. The other direction (category to container to category) has the same flavour: hom-sets get wrapped in a subtype over a dependent sum, equal only up to the evident bijection.

6 Morphisms as cofunctors

Definition 6.1 (Directed-container morphism)**.** A **morphism of directed containers** $\varphi : C \Rightarrow D$ consists of a shape map $f : \mathrm{Shape}(C) \to \mathrm{Shape}(D)$ and a contravariant position map $f^\sharp_s : \mathrm{Pos}(D)(f(s)) \to \mathrm{Pos}(C)(s)$, satisfying:

M0. $f(s \downarrow f^\sharp_s(h)) = f(s) \downarrow h$;

M1. $f^\sharp_s(\mathsf{o}(f(s))) = \mathsf{o}(s)$;

M2. $f^\sharp_s(h \oplus k) = f^\sharp_s(h) \oplus f^\sharp_{s \downarrow f^\sharp_s(h)}\big((\mathrm{M0})^{-1} \cdot k\big)$.

Definition 6.2 (Cofunctor)**.** A **cofunctor** $\phi : \mathcal{C}(C) \to \mathcal{C}(D)$ consists of an object map obj and a lift $\mathrm{lift}_s : \mathrm{Pos}(D)(\mathrm{obj}(s)) \to \mathrm{Pos}(C)(s)$, satisfying:

C0. $\mathrm{obj}(s \downarrow \mathrm{lift}_s(h)) = \mathrm{obj}(s) \downarrow h$;

C1. $\mathrm{lift}_s(\mathsf{o}(\mathrm{obj}(s))) = \mathsf{o}(s)$;

C2. $\mathrm{lift}_s(h \oplus k) = \mathrm{lift}_s(h) \oplus \mathrm{lift}_{s \downarrow \mathrm{lift}_s(h)}\big((\mathrm{C0})^{-1} \cdot k\big)$.

Remark 6.3. C0/C1/C2 are M0/M1/M2 read through the dictionary of $\mathcal{C}(-)$: sub-shape is codomain, root is identity, shift is composition. The two structures carry identical data.

Definition 6.4 (Correspondence)**.** Every directed-container morphism φ determines a cofunctor $\varphi^\wedge$ by reading $(f, f^\sharp)$ as $(\mathrm{obj}, \mathrm{lift})$. Every cofunctor ϕ determines a directed-container morphism $\phi^\vee$ by the inverse reading.

Theorem 6.5 (Hom-set bijection)**.** *The two assignments are mutually inverse:*

$$(\varphi^\wedge)^\vee \;=\; \varphi, \qquad (\phi^\vee)^\wedge \;=\; \phi. \tag{3}$$

Consequently there is a bijection between directed-container morphisms $C \Rightarrow D$ and cofunctors $\mathcal{C}(C) \to \mathcal{C}(D)$.

Proof. By lean. □

7 The category of cofunctors

Definition 7.1 (Identity and composition)**.** The **identity cofunctor** on $\mathcal{C}(C)$ is the identity on objects and on positions. The **composite** of cofunctors $\phi : \mathcal{C}(C) \to \mathcal{C}(D)$ and $\psi : \mathcal{C}(D) \to \mathcal{C}(E)$ has object map $\mathrm{obj}_\psi \circ \mathrm{obj}_\phi$ and lift

$$m \;\mapsto\; \mathrm{lift}_\phi(s,\, \mathrm{lift}_\psi(\mathrm{obj}_\phi(s),\, m)).$$

Lemma 7.2 (Composite transport)**.** *Let $\psi : \mathcal{C}(D) \to \mathcal{C}(E)$, equalities $e : y = x$ in $\mathrm{Shape}(D)$ and $e_2 : \mathrm{obj}_\psi(x) = z$, and $k \in \mathrm{Pos}(E)(z)$. Then*

$$e^{-1} \cdot \left(\mathrm{lift}_\psi(x,\, e_2^{-1} \cdot k)\right) \;=\; \mathrm{lift}_\psi\big(y,\, \left((\mathrm{obj}_\psi \cdot e) \cdot e_2\right)^{-1} \cdot k\big). \tag{4}$$

Proof. By lean. □

Theorem 7.3 (Cofunctors form a strict category)**.** *Identity and composition of cofunctors satisfy left unit, right unit, and associativity strictly. Directed containers and cofunctors therefore form a strict category* **Cof**.

Proof. By lean. □

Definition 7.4 (Operations in **DCont**)**.** Identity and composition of directed-container morphisms are obtained by transporting the cofunctor operations across the hom-set bijection of Theorem 6.5.

Theorem 7.5 (Isomorphism of categories)**.** *The assignment $\varphi \mapsto \varphi^\wedge$ preserves identities and composition, is the identity on objects, and is bijective on hom-sets. Hence it is an isomorphism of categories*

$$\mathbf{DCont} \;\cong\; \mathbf{Cof}. \tag{5}$$

Proof. By lean. □

8 Discussion

What has been established. Directed containers and small categories are intertranslatable at the object level, inverse up to a canonical position bijection. At the morphism level the translation is tighter: directed-container morphisms and cofunctors are the same data, and the categories they form are isomorphic.

Why cofunctors rather than functors. Container morphisms pull positions backward. Functors push morphisms forward. The mismatch is definitional; cofunctors (object map forward, morphism action backward) are the matching notion.

The remaining gap. Upgrading the pointwise position bijection to an equivalence of categories $\mathbf{DCont} \simeq \mathbf{Cat}$ requires a notion of isomorphism (or equivalence) of directed containers compatible with root, sub-shape, and shift. The present development stops short of that packaging; the bijection itself is fully proved.

Machine checking. Every definition and theorem statement above corresponds to a declaration in a machine-checked formalisation. The phrase "by lean" indicates that the proof has been verified by a proof assistant.

Acknowledgements

The mathematical content follows Ahman and Uustalu, *Directed containers as categories* (2016), together with the cofunctor reading of directed-container morphisms. The presentation here is a prose rendering of a formal development intended for human review of the theorem statements.

References

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